

Task 5 — Exploratory Data Analysis (EDA)

Dataset: synthetic customer-orders data generated for this learning task. It is illustrative and does not represent a real business.

1. Objective and method

This report demonstrates data inspection, descriptive statistics, missing-value and duplicate checks, plus univariate, bivariate and multivariate visual analysis using Pandas and Matplotlib.

2. Dataset overview

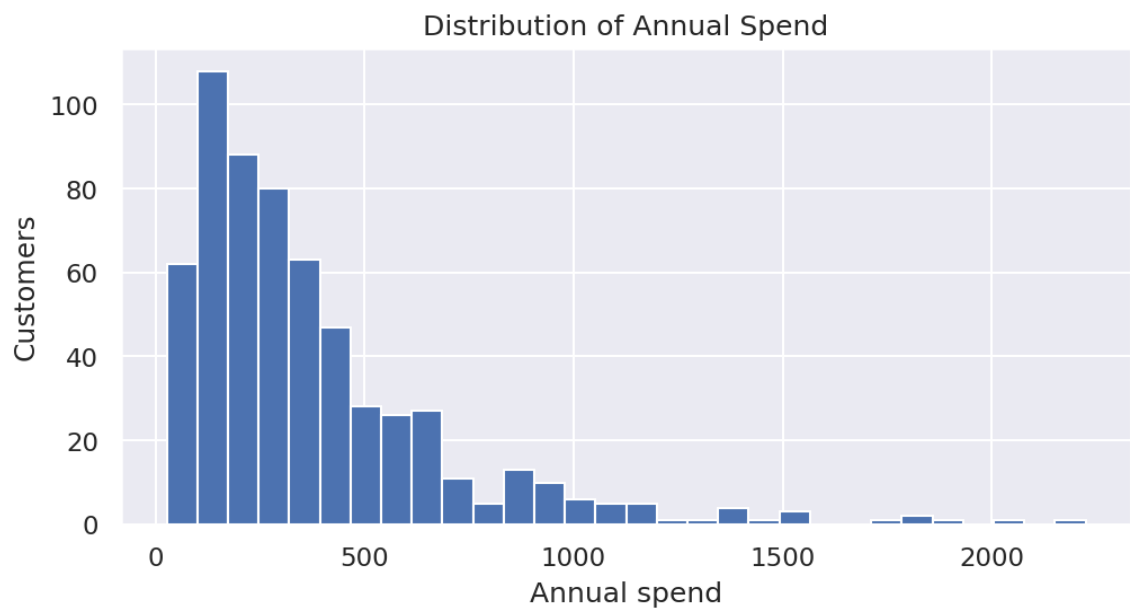
Metric	Result
Rows in raw file	604
Columns	10
Exact duplicate rows	4
Missing Age values	18
Missing Satisfaction_Score values	12
Rows after duplicate removal	600
Mean annual spend	374.53
Median annual spend	286.22
Mean orders last year	4.16
Mean satisfaction score	3.82 / 5

3. Data quality

The raw file contains 4 exact duplicate rows. Age has 18 missing values and Satisfaction_Score has 12 missing values. The analysis removes exact duplicates and fills these two numeric columns with their median values. Real projects should investigate missingness before selecting a handling strategy.

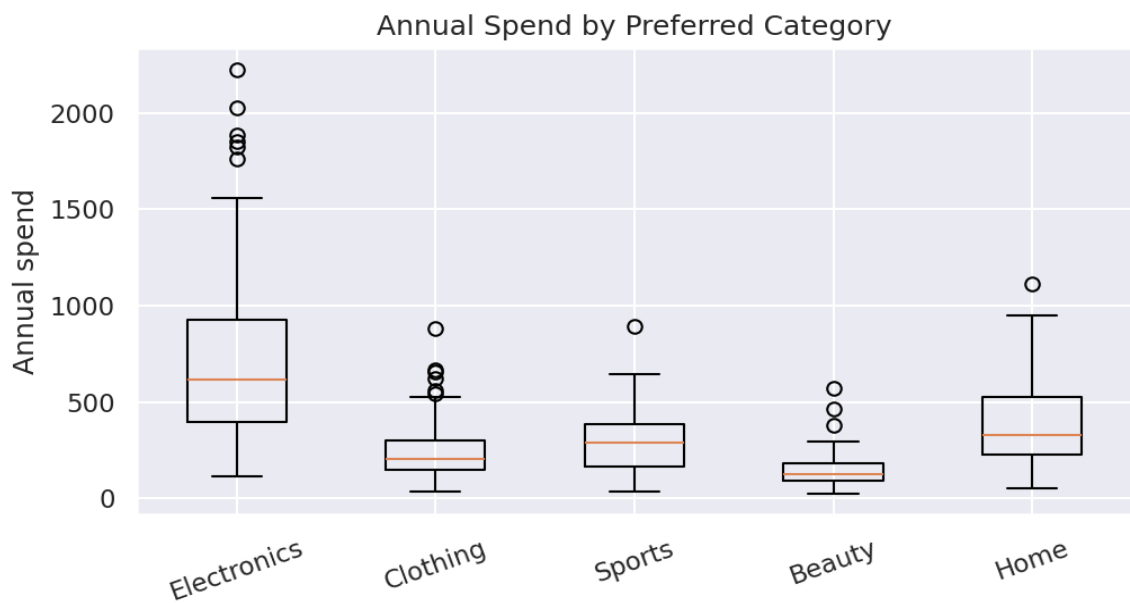
4. Visual analysis

Annual spend distribution



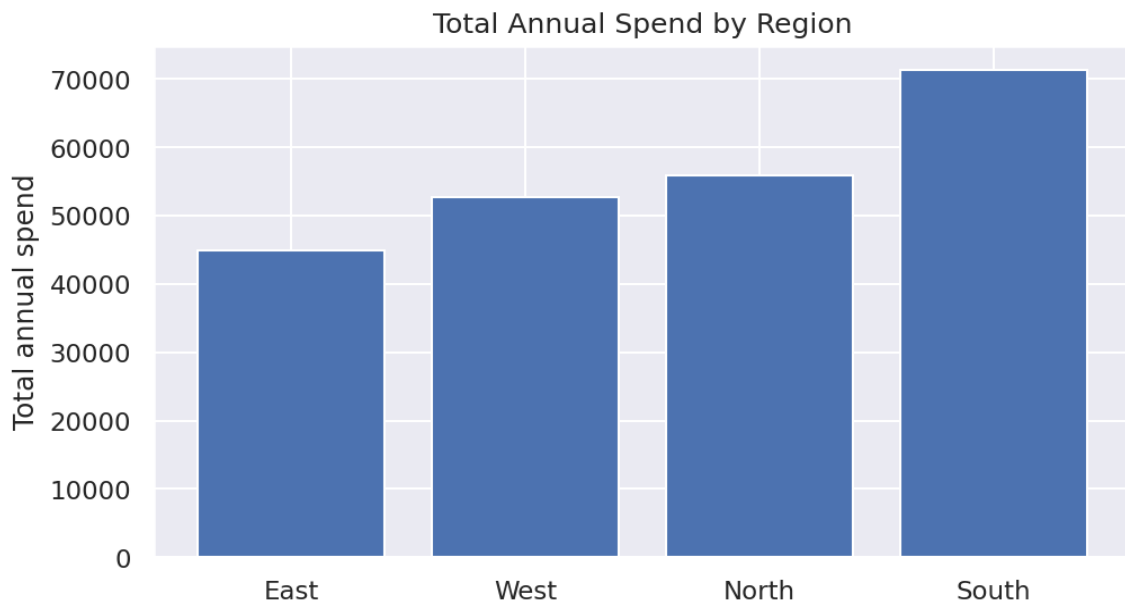
The histogram helps assess skewness and identify unusually high spending.

Spend by preferred category



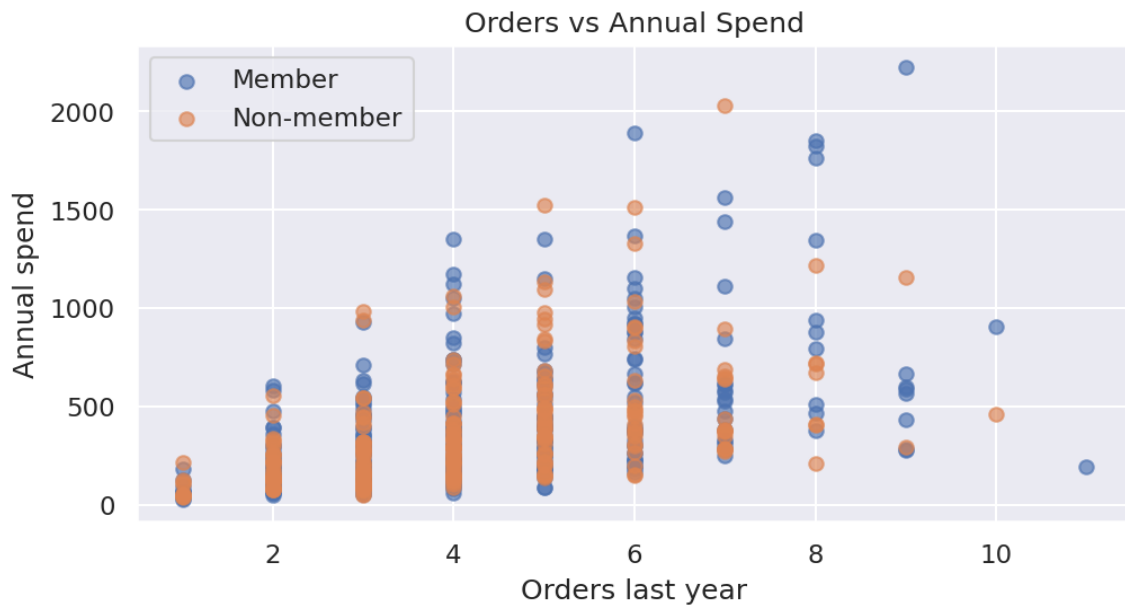
The boxplot compares medians, spread and potential outliers across categories.

Total spend by region



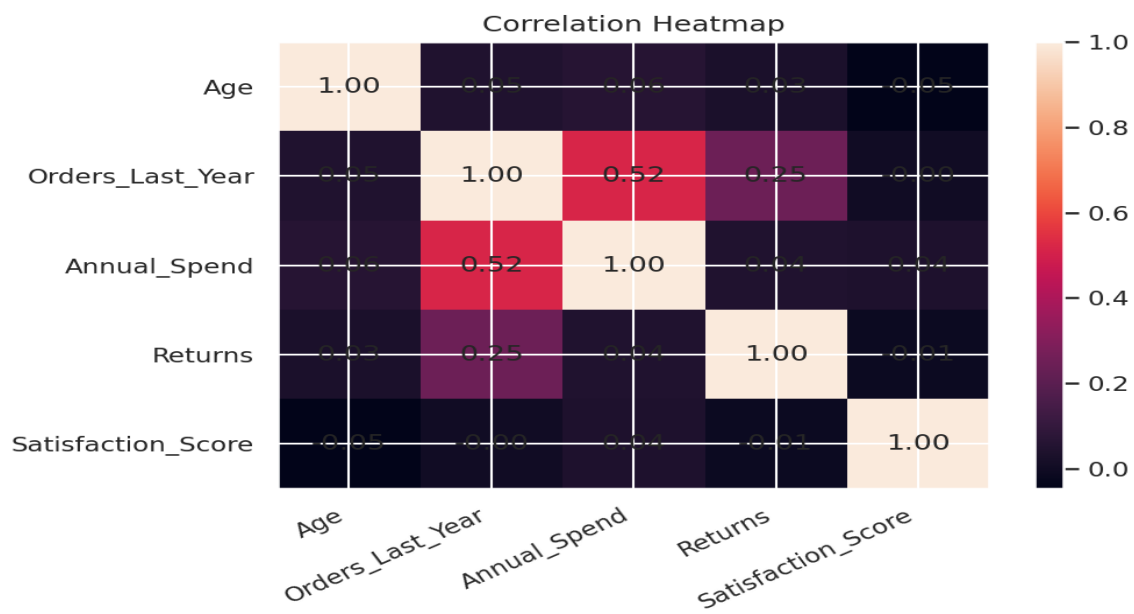
The chart compares aggregate annual spend across regions.

Orders versus annual spend



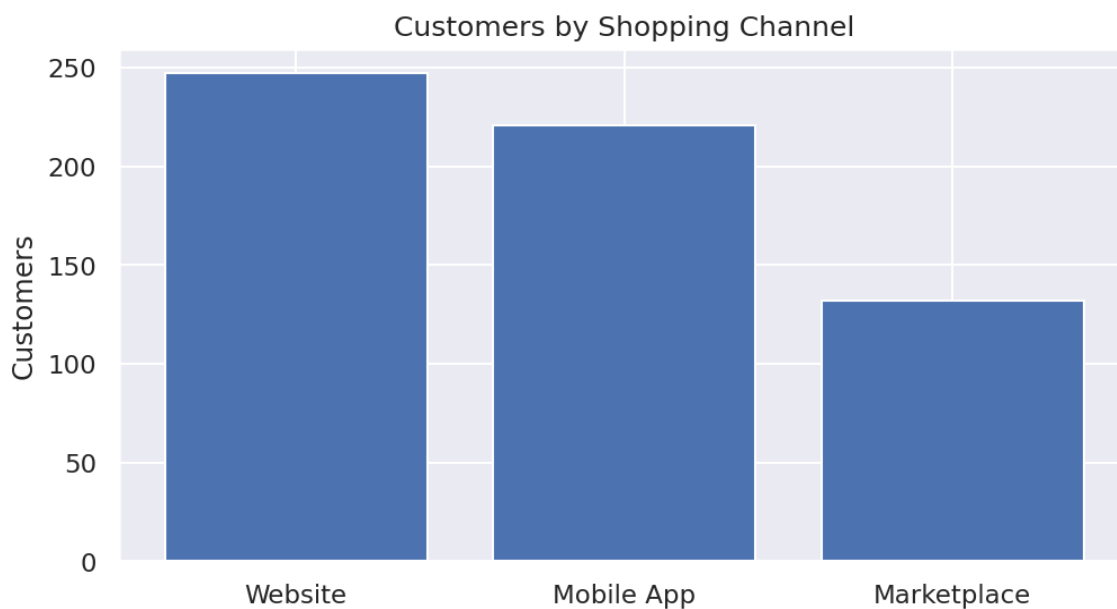
The scatterplot explores the relationship between order frequency and spend.

Correlation heatmap



The heatmap summarizes linear relationships among numeric columns; correlation does not prove causation.

Customers by channel



The chart compares the number of customers in each channel.

5. Summary of findings

- Cleaned analysis rows: 600.
- Mean annual spend: 374.53; median: 286.22.
- Mean order count: 4.16.
- Most common shopping channel: Website.
- Highest total spend region in this sample: South.
- Highest average spend category in this sample: Electronics.

6. Limitations

This dataset is synthetic and intended only to demonstrate EDA techniques. The results should not be generalized to real customers or organizations. In a real analysis, validate the data source, investigate outliers with domain experts and confirm findings before making decisions.